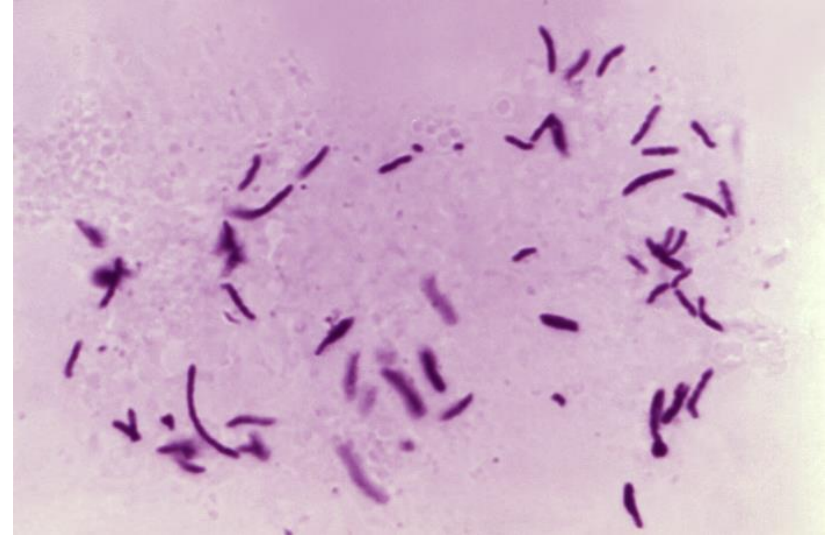
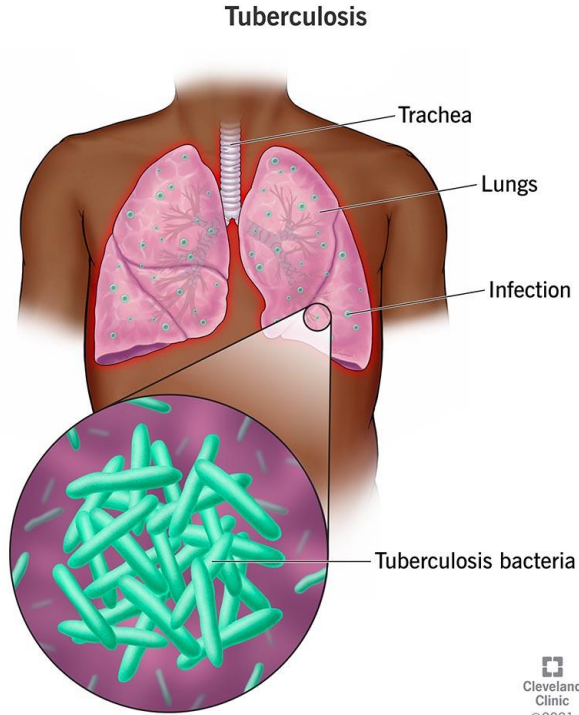


Discovering Novel *Mycobacterium Tuberculosis* (Mtb) Mutations

By Nabil Islam

Supervised Dr. Michael Hall and Prof. Lachlan Coin

What is Tuberculosis (TB)?

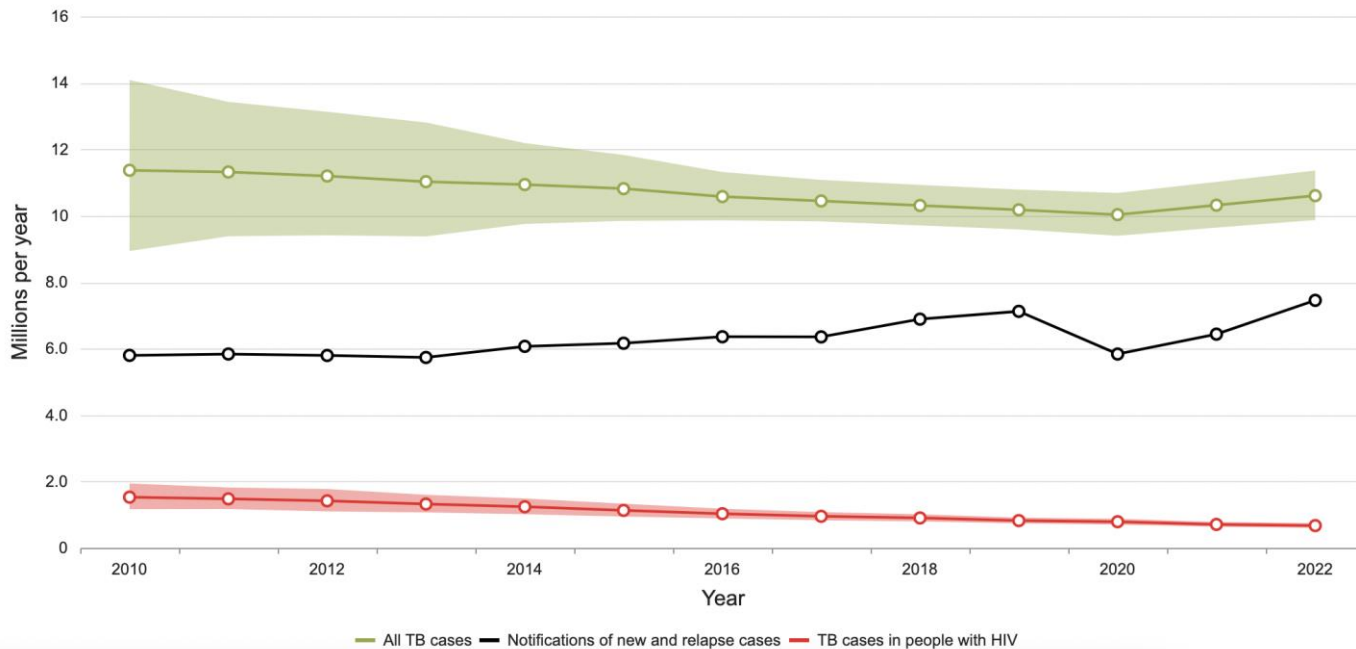


Global Impact of TB

Fig. 1.1.1 Global trends in the estimated number of incident TB cases (a) and the incidence rate (b), 2010–2022

Shaded areas represent 95% uncertainty intervals. The horizontal dashed line shows the 2025 milestone of the End TB Strategy, which is a 50% reduction in the TB incidence rate between 2015 and 2025.

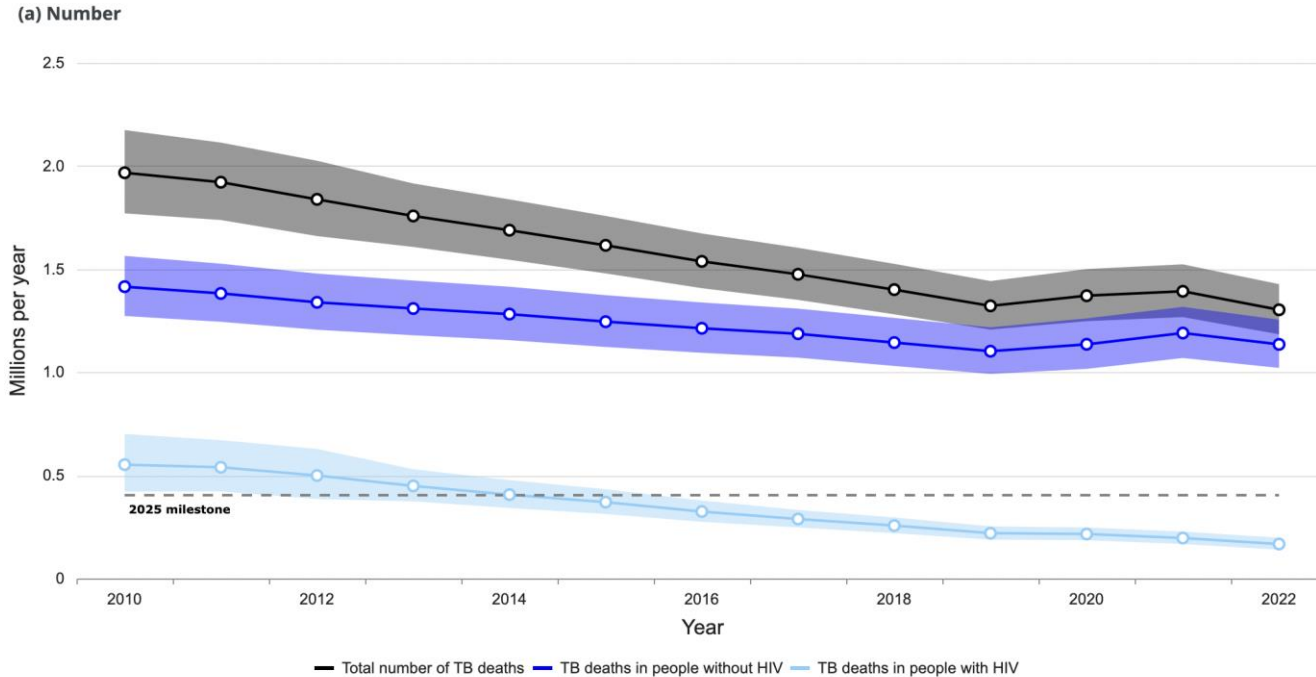
(a) Number



Global Impact of TB

Fig. 1.2.1 Global trends in the estimated number of deaths caused by TB (a) and the TB mortality rate (b), 2010–2022

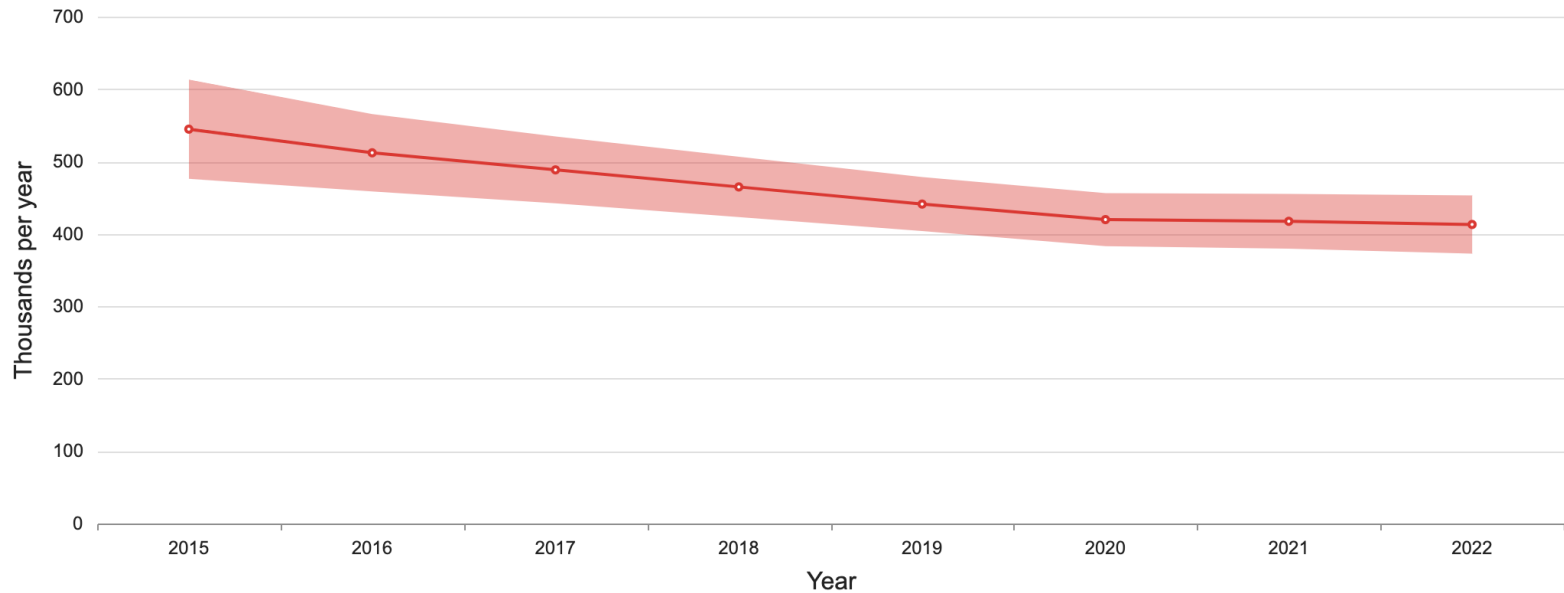
Shaded areas represent 95% uncertainty intervals. The horizontal dashed line shows the 2025 milestone of the End TB Strategy, which is a 75% reduction in the total number of deaths caused by TB between 2015 and 2025.



Drug Resistance: A growing issue

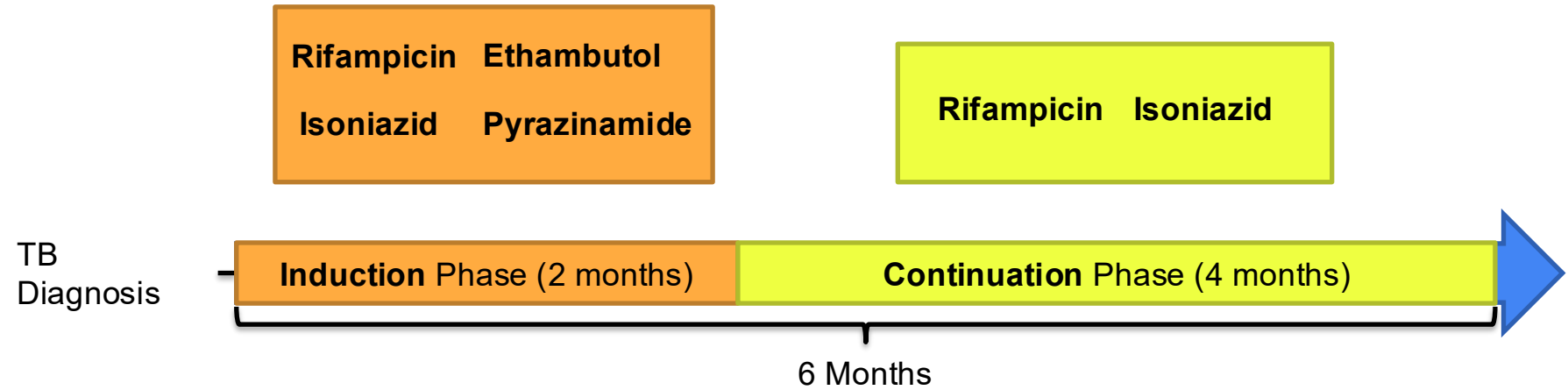
Fig. 1.3.1 Global trend in the estimated number of people who developed MDR/RR-TB (incident cases), 2015–2022

The shaded area represents the 95% uncertainty interval.

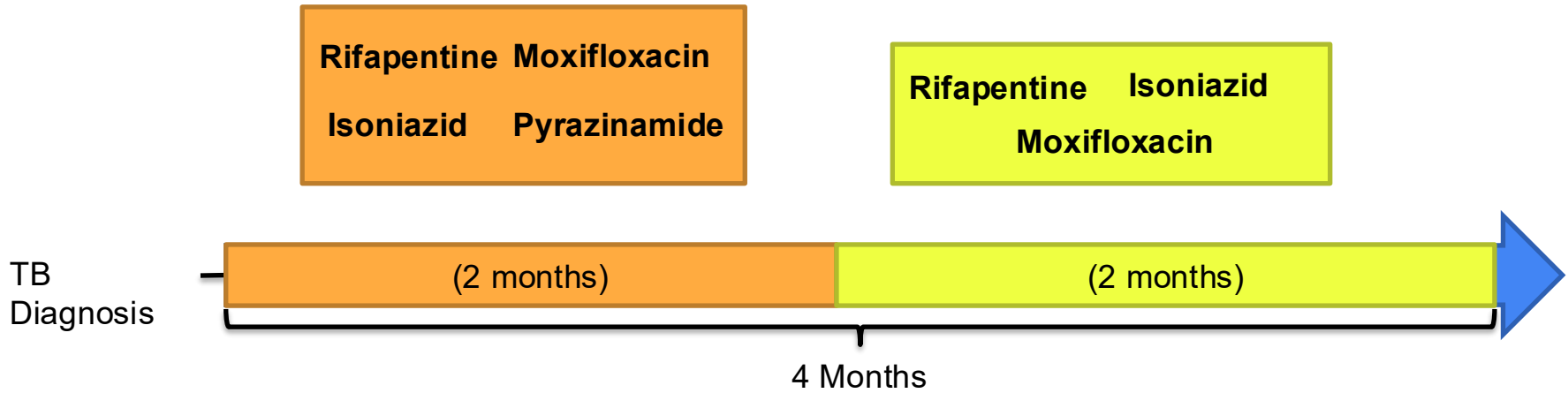


Types of Drugs to Treat TB

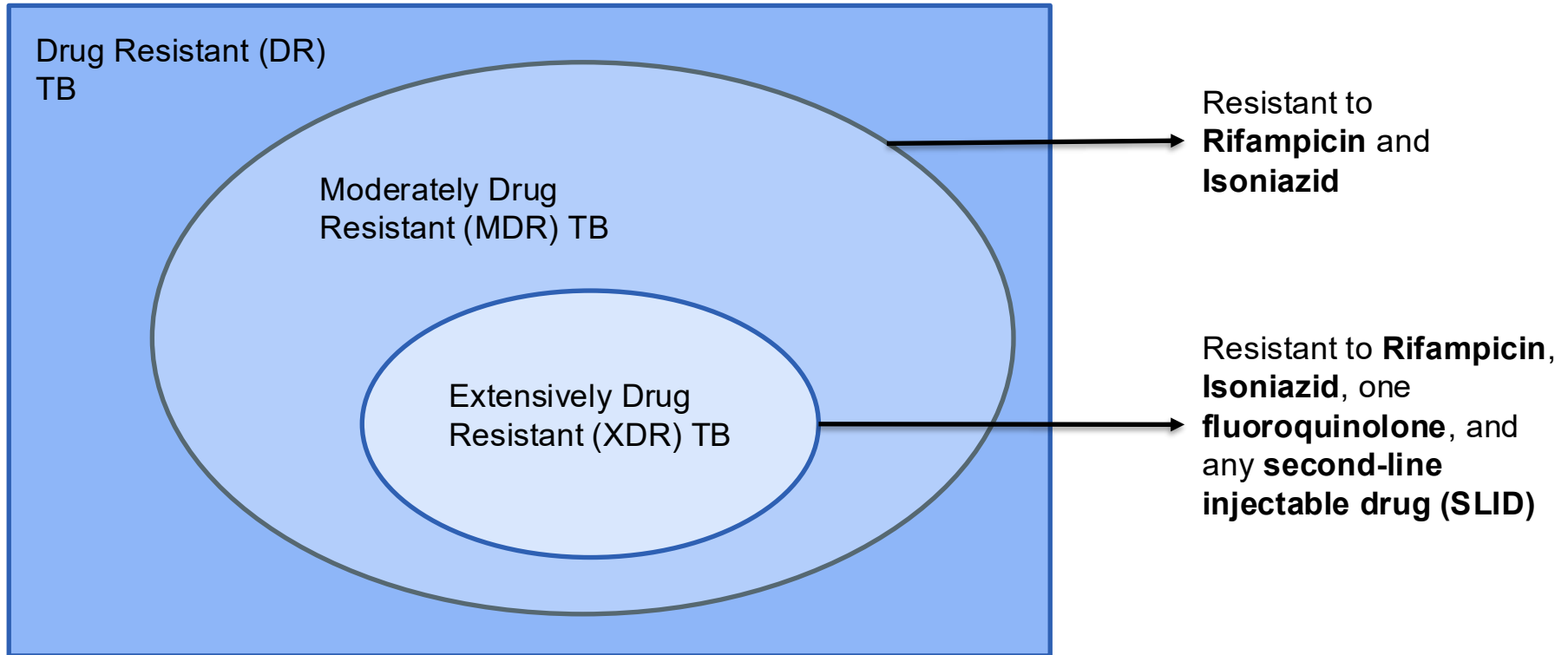
Treating TB - Drug Susceptible



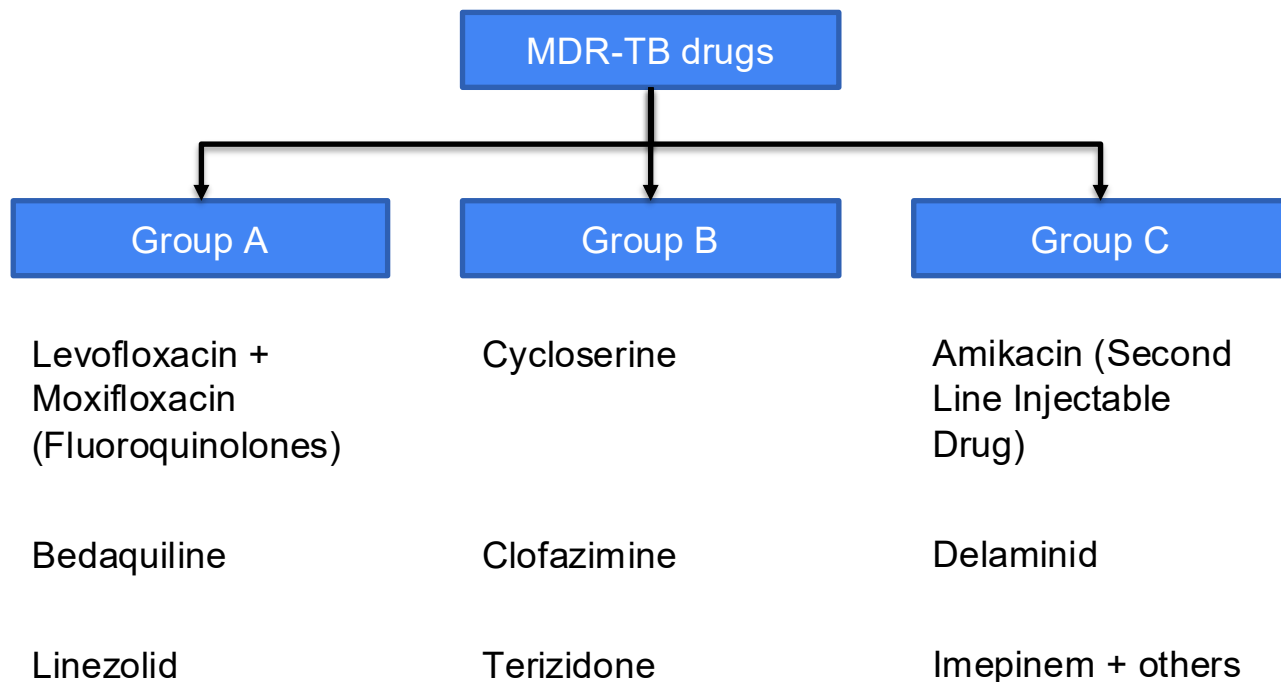
Treating TB - Drug Susceptible



Types of Drug Resistance



Treating TB - Drug Resistant



🕒 18 – 20 Months

Regimen:

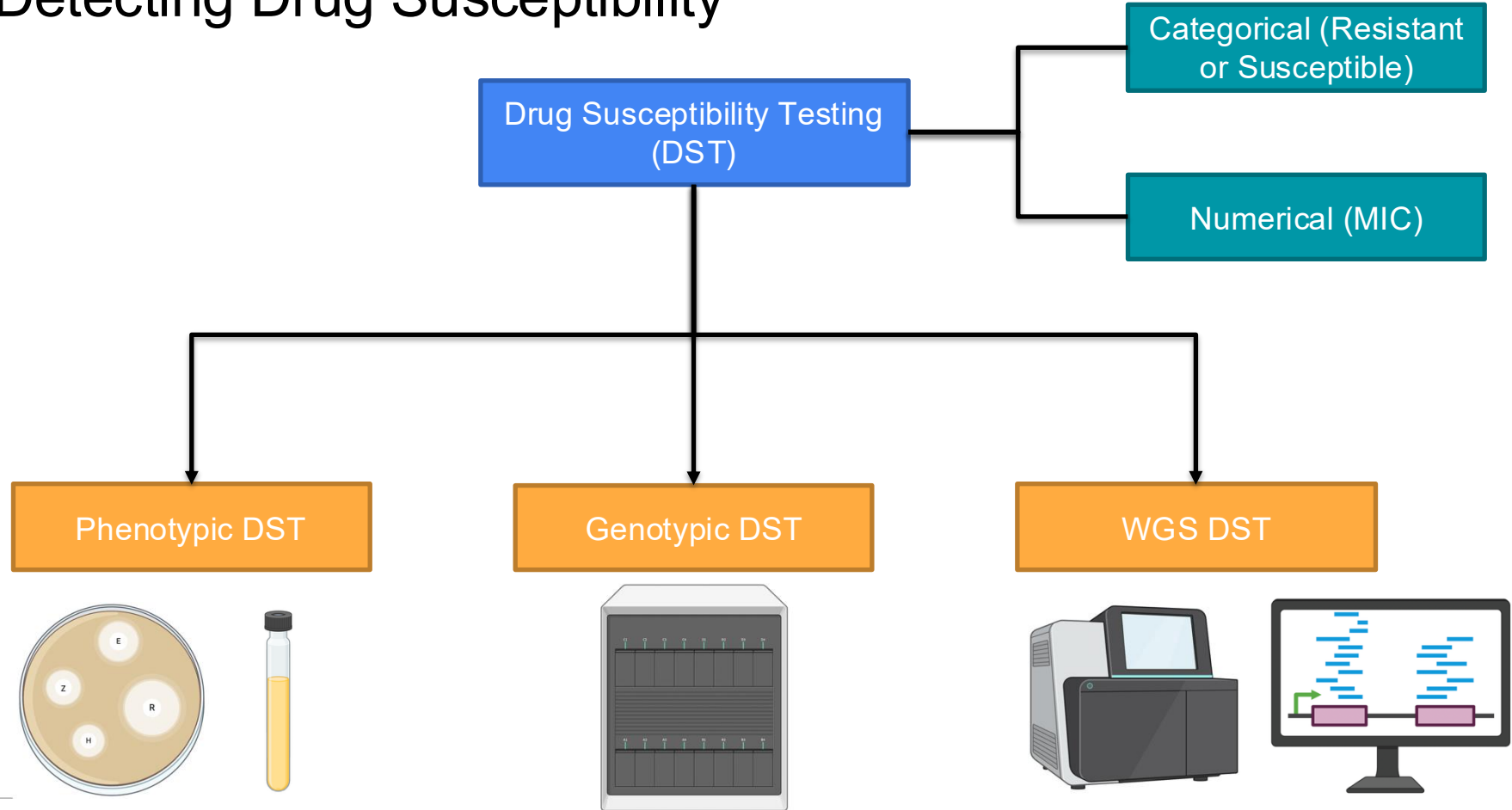
- All 3 Group A drugs and 1 Group B
- Add Group C if necessary

Treating MDR Alternatives

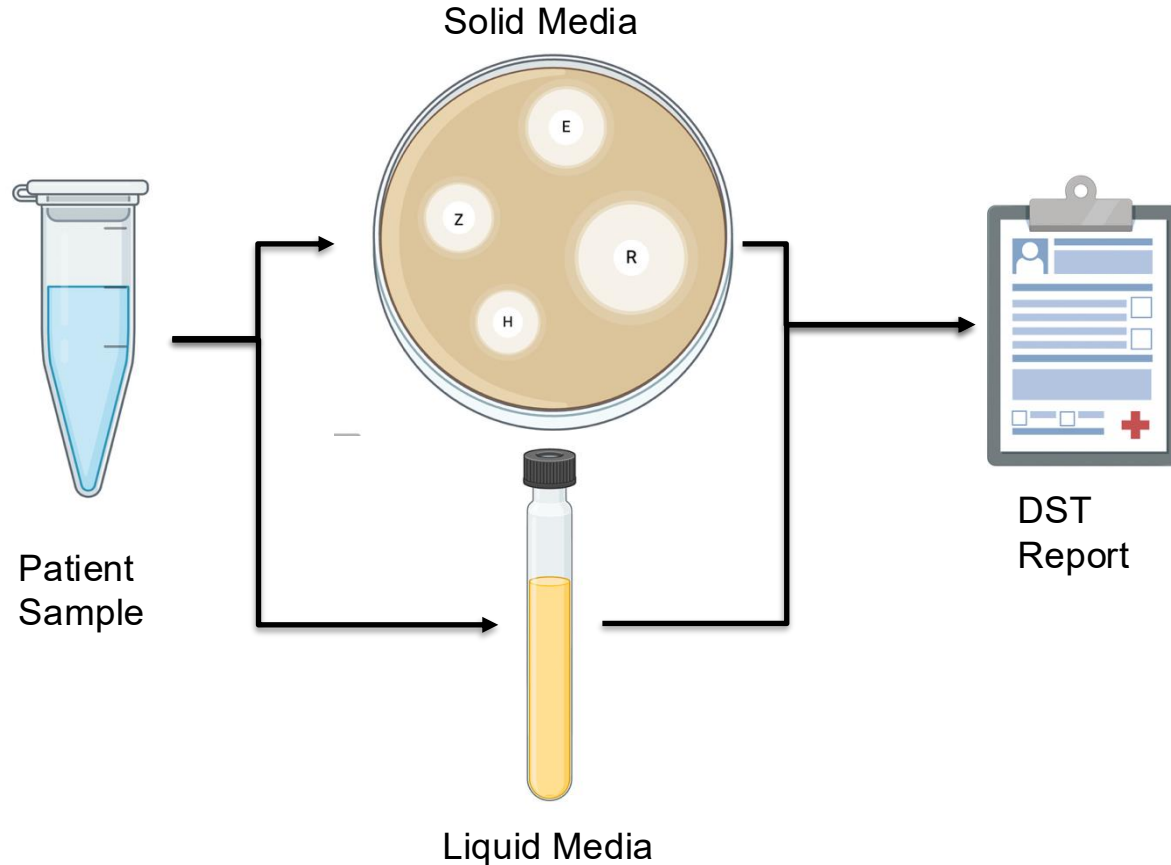
- **New WHO regimens:**
 - **6-month BPaLM regimen:** Bedaquiline, Pretomanid, Linezolid, Moxifloxacin.
 - **9-month oral regimen** under conditional recommendation

Detecting Drug-Resistance in TB

Detecting Drug Susceptibility



Phenotypic Drug Susceptibility Testing (pDST)



🕒 14 – 42 days.



Direct



Thorough

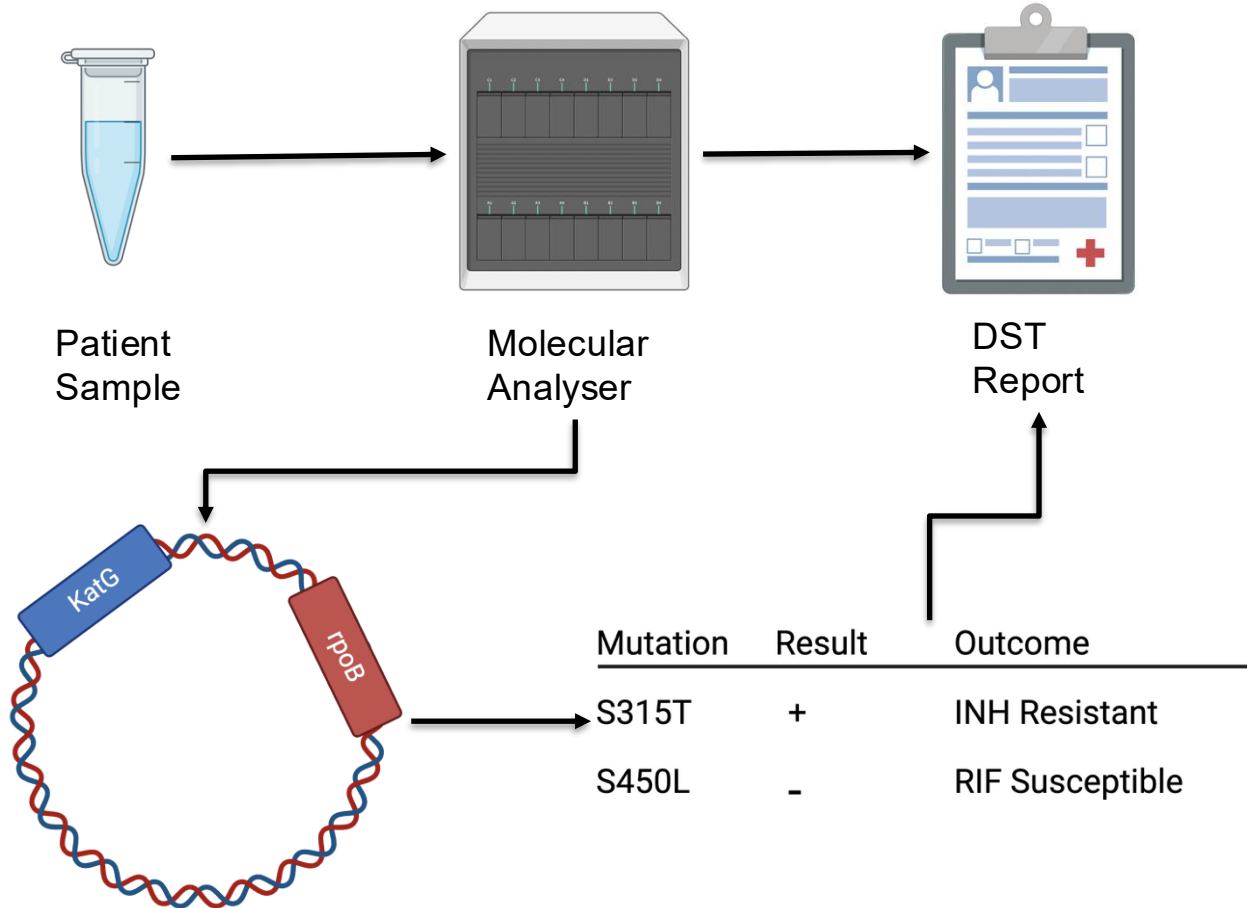


Slow



Demanding

Genotypic Drug Susceptibility Testing (gDST)



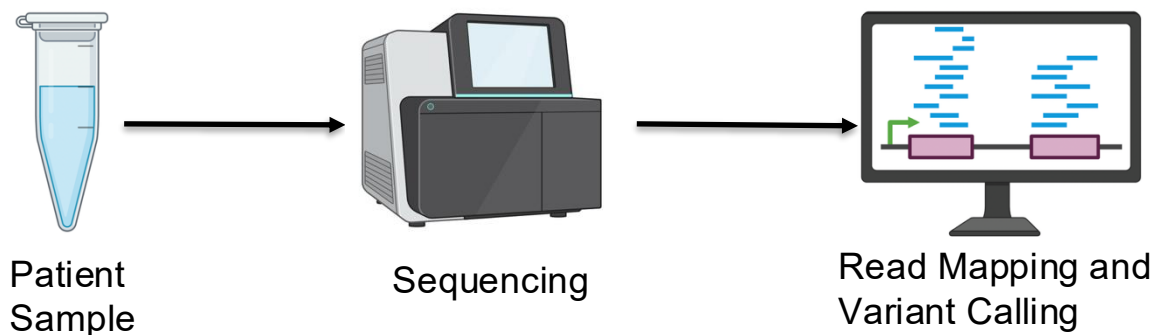
⌚ ~ 2 hours

✓ Rapid

✓ User-Friendly

✗ Limited Scope

✗ Prone to Inaccuracies

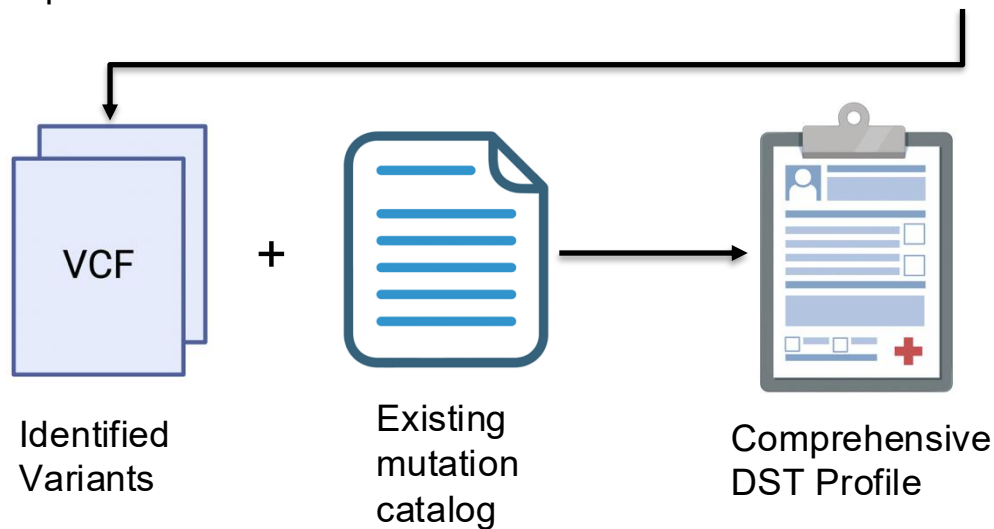


🕒 ~ 7 – 14 days

✅ Comprehensive

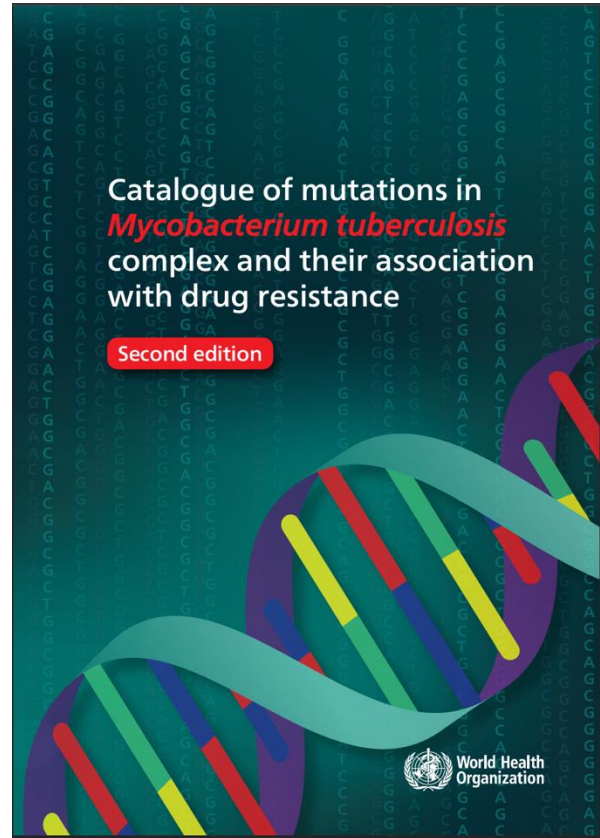
✅ Fast

❌ Incomplete Coverage



The WHO Catalogue of Mutations

- **Purpose**
 - **International standard** for *interpreting* **genotypic** data for **DST**
 - *Supports* **national TB programs** and **diagnostic developers** – *improves:*
 - **treatment decisions** thus **patient outcome**
- **Scope**
 - *Intentionally* **Non-exhaustive**
 - Only mutations with **high confidence** included



Addressing the Issue – Mtb Genome Masking

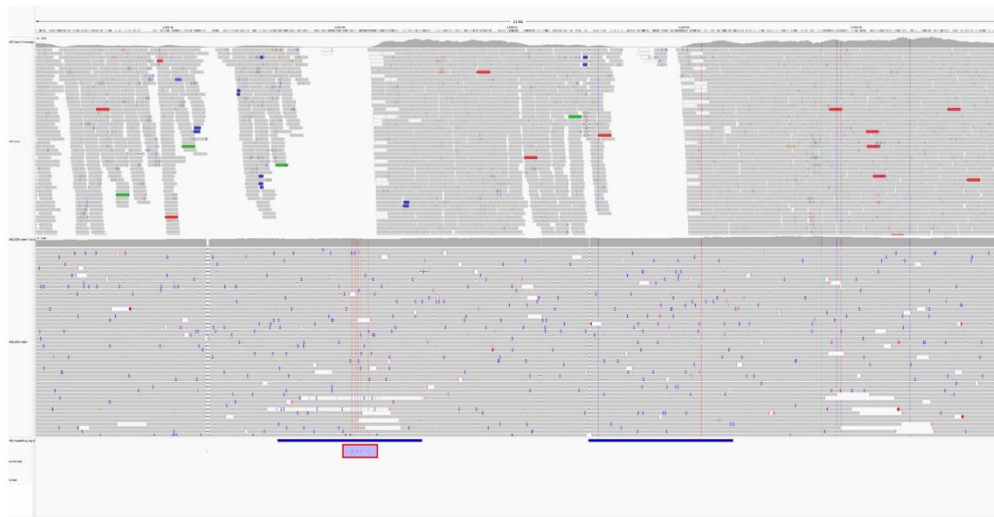
Masking Practices of the Mtb Genome

- Current Catalogues – *Non-Exhaustive*
- **Why?** – Reference Genome (H37Rv) *Masking*
 - Putative Low Confidence (PLC) Regions (Marin et al., 2022)
 - PE/PPE Gene Family
 - Repetitive or Low Complexity Regions
 - Mobile Genetic Elements

Permissive Masks – Sequencing Challenges

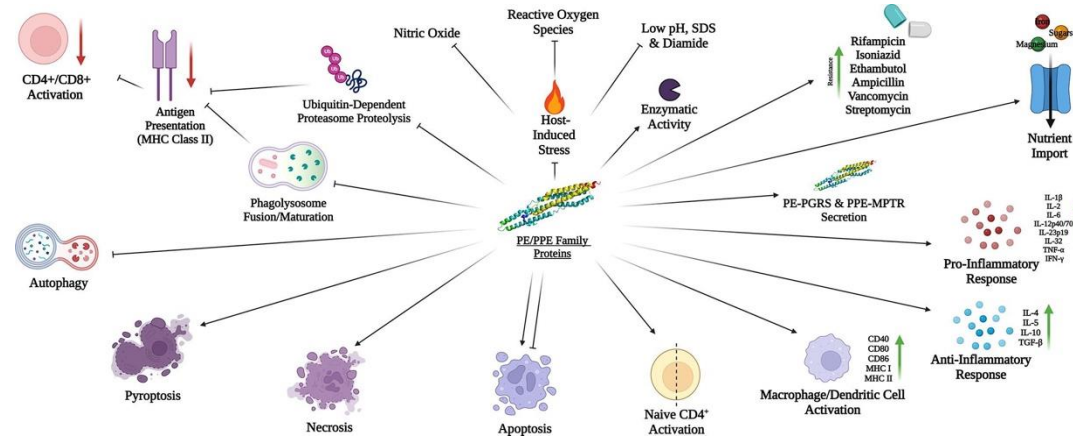
- **Short-read sequencing:**

- Limited alignment to **repetitive/homologous regions**
- Leads to **missed or poorly reconstructed regions** .
- Causes *erroneous* **Variant Calls**



PE/PPE Proteins - Immune Modulation

- 168 Genes
- *Involved* in Immune Modulation, Inflammatory Signalling, Cell Wall Changes
- **PE** or **PPE** motif in **N-terminal**
- *Highly* variable **C-terminal**



D'Souza et al., 2023

PE/PPE – Drug Resistance

- **PE-PPE genes** complicate treatment by introducing **additional resistance layers**
- **katG-PPE54**: Resistance to **isoniazid (INH)**.
 - katG mutation + PPE54 expression impairs drug activation.
- **rpoB-PPE54**: Resistance to **rifampicin (RIF)**.
 - Mutations disrupt RNA polymerase, reducing drug efficacy.
- **embA-PPE68** and **embB-PPE54**: Resistance to **ethambutol (EMB)**.
 - Mutations alter cell wall synthesis, reducing drug action

Permissive Masks – Masking, Mappability, and PE/PPE Genes

- Marin et al. - **Refined low-confidence regions (RLC)** – *permissive* mask
 - **Refined** and **Targeted** Mask – informed *empirically*
 - Covers only **4%** instead of **10%** of genome
 - Maintained **Precision** with sacrificing **recall**
 - **89.2% of false positives** occur in 1.5% (65 kb) of the genome, requiring better filters.
- **PE/PPE genes:**
 - **52 of 168 PE/PPE genes** have *high* mappability scores.
 - **PE_PGRS** and **PPE_MPTR** subfamilies show **poor mappability** and high sequencing errors.

Aims and Scope of the project

- **What we have:** Phenotypic resistance data (for at least 1 drug) in 45,000 isolates
- **Aim: Data mining** for novel mutations, especially in **previously masked regions**
 - Focus on the **PE/PPE** candidate gene family
 - Use permissive masks
- **Plan for early analysis:** False Negatives (Phenotype is Resistant but No Linked Mutation)

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